KRITIK SOMAN

Applied AI/ML Engineer with 6+ yrs. experience

Email: kritiksoman@gmail.com | GitHub: github.com/kritiksoman | LinkedIn: https://www.linkedin.com/in/kritik-soman/ | Portfolio: https://kritiksoman.github.io/online-cv/

PROFESSIONAL EXPERIENCE

Loblaw Digital (via Hays)

Machine Learning Engineer (Sept. 2025 - Present)

Gatsby: Conversational e-commerce agent
•Implemented in-person and online order history tool, orchestration to plan meals based on context created with store information, user profile, dietary restrictions and purchase history.

Accenture

AI/ML Computational Science Associate Manager (Sept. 2025)
AI/ML Computational Science Specialist (April 2022 - Aug. 2025)

ApprovalX: Intelligent Compliance Automation Platform
• Developed a **multi-lingual agentic AI** system in **python** using **LangGraph** and **GPT-4o** to automate review workflows for marketing assets like emails and banners. Built **orchestration agent, branding agent, functional agent**, **compliance agent, and reporting agent**. Deployed with **Ray serve, FastAPI, Kafka, and Docker.**
• Performed **SFT, PEFT on Qwen 2.5-VL** for performing various compliance checks on documents.
• Delivered a **30% reduction in Average Handling Time (AHT)** for manual QA through automation and intelligent document parsing.

InstructGen: GenAI-Powered Marketing Automation Suite
• Built a Retrieval-Augmented Generation (**RAG**) pipeline in python for brand specific email **HTML generation** with custom CSS.
• Developed a **RAG**-based **SQL generation** engine in python for customer targeting by merging sub-queries indexed in a vector database.
• Achieved a **50% reduction in AHT** for marketing email creation and campaign targeting through end-to-end automation.

Xtracto: Cross-Format Document Parsing Engine
• Architected a standardized parser in python for diverse digital formats (PPTX, PSD, PDF, HTML, raster images), integrating tools like pdfminer, psd-tools, Aspose API, Playwright, pdf2image, Google OCR, Microsoft Florence VLM, and HiSAM for LLM-based automation.

Leadership & Mentorship:
• Leading a team of 4 AI Analysts through full-cycle development, driving iterative improvements and production readiness.
• Conducted hiring and technical interviews for entry-level AI Analyst roles, contributing to team growth and talent acquisition.

App Orchid

Junior Data Scientist (August 2020 - April 2022)

• **NLP**: Created methods for extraction of custom entities in contracts based on Question Answering (QA), RegEx, Shallow Syntax Parsing, Token REGEX, Relative Time Expression, Language Identification, Fine tuning QA, Sentence similarity models, Classification of Document, Clauses.
• **CV**: Row and column detection in table images using cascade RCNN. Heading, text, table detection in contract images using YOLO v3+.

STMicroelectronics

Intern (February 2016 – June 2016)

Undergraduate thesis on Motion Algorithms using MEMS Sensors.

PATENTS / PUBLICATIONS

• Soman, Kritik. “Agentic Numeric Optimization for Nutrition Planning” In ICML 2026 AI4Math Workshop Poster.
• Method and System for Modifying Images (US Patent No: 20260148577)
• Artificial Intelligence based System and Method for Document Analysis (US Patent Filed)
• Soman, Kritik. “Rooftop Detection using Aerial Drone Imagery.” In Proceedings of the ACM India Joint International Conference on Data Science and Management of Data, pp. 281-284. 2019.

EDUCATION

M.Tech. in *Machine Learning*
Indian Institute of Technology, Kanpur
August 2018 – June 2020 | GPA: 9.3/10
• Thesis on **Semantic Mobile Base Station Placement**: Developed a method using Genetic Algorithm and Semantic Segmentation for placement of mobile base stations at optimal location given RGB and depth aerial images. (Supervisor: Prof. K.S. Venkatesh)

B.Tech. in *ECE* – JIIT Noida
August 2012 – June 2016 | GPA: 9/10

OxML 2026 – Attended 1 week in person Summer School held at Oxford Mathematical Institute, U.K.

AWARDS

Accenture Technology Wizard Award 2025 | Excellent Mentor 2023 | Client Value Creation Award 2023 | IIT Kanpur Academic Excellence Award for 2018-19 |
ACM CoDS-COMAD Travel Grant 2019, 2020

OPEN SOURCE

• **GIMP-ML** - Deep Learning Models for GNU Image Manipulation Program (1.6k ⋆ on GitHub), News: Visual Informatics Group @ UT Austin, Papers With Code (Top Trending), Presented at PyCon India 2024 @Google

FRAMEWORKS

Python | **AI Agents** | **LangGraph** | **RAG** | **LLMs** | Google Agent Development Kit | **PyTorch** | **Transformers** | Spacy | OpenCV | skimage | Pandas | playwright | psd tools | Postgres | FastAPI | pdfminer | Git